MILITARY SPECIFICATION

BATTERIES, NON-RECHARGEABLE, LITHIUM THIONYL CHLORIDE

This specification is approved for use by U.S. Army Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers non-rechargeable batteries of the non-reserve type composed of electrochemical cells utilizing a Lithium Thionyl Chloride system (see 6.2).

1.2 Classification.

1.2.1 Type designation. The type designation of Lithium Thionyl Chloride non-rechargeable batteries shall be in the following form. (see 3.1).

BA-	6516	/U
Component (1.2.1.1)	Battery Type number (1.2.1.2)	Installation indicator (1.2.1.3)

1.2.1.1 Component. Non-rechargeable batteries are identified by the two-letter symbol "BA" followed by a hyphen.

1.2.1.2 Battery type number. The battery type number identifies the basic design of the battery (See 3.1) and consists of a four digit number in the 6001 through 6999 series.

1.2.1.3 Installation Indicator. The installation indicator identifies equipment the battery is used in; i.e. /PRC-25 or if "Universal", i.e. /U indicates use in various equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SLCET-R, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6135

AMSC N/A

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2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issures of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-571	Solder, Tin Alloy; Tin-Lead Alloy; amd Lead Alloy

MILITARY

MIL-W-76	Wire and cable, Hook-up, Electrical, Insulated.
MIL-P-116	Preservation, Methods of.
MIL-W-6858	Welding, Resistance, Spot and Seam.
MIL-F-14072	Finishes for Ground Electronic Equipment.
MIL-F-14256	Flux, Soldering, Liquid (Rosin Base).
MIL-B-43014	Boxes, Water Resistant Paperboard, Folding, Set-up and Metal-stayed.

STANDARDS

FEDERAL

FED-STD-101	Test Procedures for Packaging Material
FED-STD-595	Colors.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-1360	Fuses, Fuseholders and Associated Hardware, Selection of.

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MIL-STD-45662 Calibration System Requirements.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specifciation to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on the date of the solicitation.

SPI 1G00066	Special Packaging Instruction
DOT-E 7052	Transportation Exemption for Lithium Batteries

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Selection of specification and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 and shall be approved by the procuring activity.

3.3 First article. When specified in the contract or purchase order, samples shall be subjected to first article inspection (see 4.6 and 6.2).

3.4 Materials. Materials listed in Table I, when used, shall be in accordance with the applicable specification (see 4.4 and 6.2).

3.4.1 Metals. All metals which do not enter into the basic electro-chemical reaction of the cell shall resist, or be treated to resist corrosion in accordance with MIL-F-14072.

3.4.1.1 Dissimilar metals. When dissimilar metals which would adversely affect battery performance are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided in accordance with MIL-F-14072.

3.4.2 Insulating compounds - flow or shrinking. When tested as specified in 4.8.14 the insulating, impregnating, potting and sealing compounds shall no flow at high temperature, nor crack or draw away from the sides of a container at low temperature. Any compound used shall be non-flammable and non-toxic. Potting shall not inhibit cell vent operations (see 4.8.10).

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3.5 Design and construction. Batteries shall be of the design, construction, physical dimension, weight, and polarity as specified in 3.1.

3.5.1 Intercell connections. Intercell connections shall be spot welded in accordance with MIL-W-6858, Class B. Connections between cell block and terminal shall be so insulated and positioned as to avoid contact with other conducting material and/or jacket of the battery. When insulated wire is soldered to terminal lugs, it shall not be bared more than 3/32 inch from the lug nor shall it extend more than 3/32 inch beyond the lug (see Table II and 4.8.18).

3.5.2 Age of cells and batteries. The manufacturer shall certify that the age of the cells and batteries are as follows: (see 6.2)

a. Minimum age of cells, from the time of their fabrication to the time of their presentation for acceptance inspection as batteries, shall be 5 days.

b. Maximum age of cells, from the time of their fabrication to the time of their shipping date, shall be 180 days.

c. Batteries shall be submitted for acceptance inspection not more than 120 days prior to shipping date (see 4.5).

3.5.3 Terminals (insulation resistance). Terminals shall be as specified on the applicable specification sheet (see 3.1) and insulation resistance shall be not less than 5 megohms when tested as specified in 4.8.8.

3.5.4 Safety features. Each cell of the battery shall contain a feature so that any potentially explosive condition caused by sustained external or internal shorting shall cause the safety feature to activate and thus preclude an explosion. This feature shall operate between 200 - 320°F. In addition to this safety precaution each complete battery shall be fused, when necessary, with a time delay fuse. If a battery type is composed of more than one section, each individual section shall be fused separately, when necessary, with a time delay fuse. The location of the time delay fuses shall be optional. Fuses shall be selected from MIL-STD-1360. Batteries shall not be constructed using parallel arrangements of cells (see 4.8.10).

3.5.5 High temperature switch. A high temperature switch or thermal fuse shall be as specified on the applicable specification sheet (see 3.1). When tested as specified in 4.8.16, the switch shall remain closed below 180°F and open at 190 ± 5°F.

3.5.6 Jackets (4.8.1) The jackets may consist of either metallic or nonmetallic material. The contents of multicell batteries shall fit snugly enough in the jackets to minimize movement of the cells. Jackets covering one or more cylindrical cells stacked end on end, and having open top and open bottom, shall be so attached to the cells as to prevent them from slipping out when held or shaken vertically.

3.5.6.1 Metallic jackets. The outside of the battery shall have a coating to protect the jacket from corrosion during or at the conclusion of any of the tests specified herein. The contents of the

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jacket shall be completely insulated from the metal unless otherwise specified. The jacket shall not open at any of its seams after being subjected to the test specified. The test shall be applicable to batteries weighing 5 pounds or more (see 4.8.11.1).

3.5.6.2 Non-metallic jackets. Non-metallic jackets, cardboard or plastic, shall not support combustion nor emit toxic vapors when subjected to flame. The jacket shall not open at any of its seams after being subjected to any test to which the battery is subjected (see 4.8.1).

3.5.6.2.1 Jacket integrity (paperboard). Paperboard jackets shall not fall apart and the seams shall remain intact when tested as specified in 4.8.11.2.

3.5.6.3 Color of Jackets. The color of exposed surfaces of jackets shall match on of the following lusterless greens 34079, 34086, 34087, 34096, 34102, 34127, and 34128 per FED-STD-595 (6.2).

3.5.7 Connectors. Connectors shall be as specified on the applicable specification sheet (see 3.1). Location of the connectors shall be verified as specified in 4.8.15. Only sockets which are electrically connected shall require pins.

3.5.8 Manufacturing processes. The contractor shall maintain records on all manufacturing processess he employs for the preparation, fabrication and/or refining of cell and battery components. These records shall be available to the Government upon request. The records shall be so maintained that any cell lot and battery fabricated from the cell lot can be identified with respect to material used, batch number, components supplier, and other pertinent process data (see 6.2).

3.5.8.1 Time order production. Battery lots shall be packaged and shipped as produced consecutively (see 6.2).

3.6 Battery voltages.

3.6.1 Battery open-circuit voltage. The open-circuit voltage shall not exceed the maximum voltage specified. (see 3.1 and 4.8.2.1).

3.6.2 Battery closed-circuit voltage. The closed-circuit voltage shall be not less than the minimum voltage specified (see 3.1 and 4.8.2.2).

3.6.3 Cell closed-circuit voltage. When cells are tested as specified in 4.8.12, the voltage shall not fall below 2.50 volts.

3.7 Capacity. When the battery is tested for capacity as specifiedin 4.8.9 the time required to reach its specified minimum voltage shall be not less than the battery capacity requirement specified 3.1).

3.7.1 Initial voltage delay. When the batery is tested for capacity, the time required at the beginning of discharge for the battery units to reach minimum voltage after the load is applied shall be not more than the time specified (see 3.1 and 4.8.9.1.1).

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3.8 Gas leakage - (cell test). Each cell used in a battery shall have a three week leakage rate not to exceed 0.005 percent of the thionyl chloride in the cell when tested in accordance with 4.8.13.

3.8.1 Low temperature discharge, (4.8.13.1). Any cell from a cell lot whose capacity deviates from the average of that cell lot by more than ±9 percent, or which is below the minimum capacity service requirement, shall be considered as failed. There shall be no venting, fire or explosion.

3.9 Vibration. After the batteries have been tested as specified in 4.8.6 they shall meet the visual and mechanical and battery voltage requirements (see 3.5 and 3.6). There shall be no voltage fluctuations during the test.

3.10 Mechanical shock. After the batteries have been tested as specified in 4.8.5 they shall meet the visual and mechanical and battery voltage requirements (see 3.5 and 3.6).

3.11 Drop test. Following the battery drop test, the socket shall remain within the limits specified on the individual specification sheet (see 3.1). The battery shall meet the electrical requirements specified. Following each drop, cells shall not be exposed and battery dimensions shall remain within tolerance (see 4.8.3).

3.12 Altitude. After the batteries have been tested as specified in 4.8.7 they shall meet the visual and mechanical and battery open circuit voltage requirements (see 3.5 and 3.6.)

3.13 Coulombic ratio. Each cell shall have a coulombic ratio (calculated based on ingredients in fabrication) of lithium to thionyl chloride that is selected by the manufacturer. The contractor shall certify that this ratio is being maintained throughout production at ± 10% for the preselected value.

3.13.1 Anode. A method of maintaining anode continuity and electrical contract shall be incorporated into anode design.

3.14 Labeling and marking. All labeling and marking shall be clear and legible throughout all the tests specified herein. Labeling and marking shall be black. Metallic and plastic jackets may have the labeling and marking engraved, or die stamped, in which case it may be the same color as the background.

3.14.1 Labels. Each battery shall have a label as specified in 3.1. If there is insufficient space to show all required information on one face of the battery, it shall be continued on another face. There shall be no information on the label other than the following:

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BATTERY, NON-RECHARGEABLE, LITHIUM THIONYL CHLORIDE
Type Designation
(Contract Number) --
(Code) -- (See Note and 3.14.1.1)
Serial Number (See 3.14.1.2)
Manufacturer's name
(Trade name may also be used)
Manufacturer's plant
DO NOT CHARGE, SHORT CIRCUIT, INCINERATE, OR MUTILATE THIS BATTERY
OTHERWISE BATTERY MAY VENT OR EXPLODE

EXAMPLE:

BATTERY, NON-RECHARGEABLE, LITHIUM THIONYL CHLORIDE
BA-6516/U
DAAB05-73-C-1234
0373B

Serial number 1-2-3
James E. Doe Company

JODOCO
Batteryville, NJ

DO NOT CHARGE, SHORT CIRCUIT, INCINERATE OR MUTILATE THIS BATTERY
OTHERWISE BATTERY MAY VENT OR EXPLODE

Note: The code may be placed on the bottom of single-cell batteries.

3.14.1.1 Code. The Code shown shall indicate the month, year and week of manufacture of the battery by means of a four-digit number followed by a single letter in which the first two digits shall indicate the year. Months earlier than the tenth month shall be a single digit preceded by "0". "A" shall be used for the first week of the month, "B" for the second week of the month, etc. Sunday shall be considered the first day of a week.

EXAMPLES:

A battery manufactured during the second week of March 1983 will bear the code "0383B".

A battery manufactured during the third week of November 1983 will bear the code "1183C".

3.14.1.2 Serial number Each battery produced and furnished under a particular contract shall be given a specific serial number. These serial numbers shall be assigned consecutively beginning with one and continuing therefrom for the entire contract lot of batteries. Serialization of batteries shall be accomplished immediately upon completion of fabrication and prior to sampling for inspection.

3.14.2 Special marking A black band, extending around approximately one third (1/3) of the battery length, shall be provided on each jacket. The term "Lithium Battery" shall appear in bold-lusterless green lettering (same as jacket color) on the black portion of the jacket on at least one surface of the battery.

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3.14.3 Terminal marking. On batteries having socket-type terminals, all markings such as polarity, voltage, and the voltage unit of battery (A, B, C, etc.) shall appear on the face of the battery bearing the socket. On other type terminals, the terminal markings may appear on the top or the side of the battery, or both. Markings shall indicate clearly the terminals to which they refer.

3.14.4 Fuse marking. Ampere rating of the fuse shall be indicated adjacent to the fuse location for replaceable fuses only.

3.15 Cell water content. Water content of the thionyl chloride electrolyte solution in a cell shall not exceed 1000 parts per million by weight when sampled after the cell has been stored for four weeks at 130°F and tested quantitatively using infrared analysis for hydrogen chloride. The contractor shall certify that this limit is not exceeded throughout production (see 6.2 and 4.8.19).

3.16 Workmanship. Batteries shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect their life, serviceability, interchangeability, or appearance in accordance with Requirement 9 of MIL-STD-454.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified in herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities shall be of sufficient accuracy, quality and quantity to permit performance of the required examinations and tests. Unless otherwise specified herein, all examinations and test shall be performed under ambient temperature, humidity, and atmospheric pressure conditions. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.1.1.1 Instrument accuracy

4.1.1.1.1 Volmeters and ammeters. All voltmeters and ammeters used in testing the batteries shall be accurate within 1 percent of the full scale value. The voltmeter and ammeter ranges shall be such that all readings are taken on the upper half of the scale. The sensitivity of voltmeters shall be not less than 10,000 ohms per volt.

4.1.1.1.2 Resistor tolerance. During all tests involving discharge through a resistance, such resistance shall be accurate within the following percentages:

	Percent
Up to and including 25,000 ohms	± 0.5
From above 25,000 ohms to and including 1 megohm	± 1.0
Above 1 megohm	± 5.0

In determining the resistance used as a test load, the resistance of all continuosly operating voltmeters shall be considered as part of the specified load.

4.1.1.1.3 Power supplies. Power supplies used for discharges specified herein shall be accurate within ±1 percent.

4.1.1.1.4 Timing. Timing equipment shall be accurate within 0.5 percent.

4.1.2 Responsibility for compliance. All items must meet all

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requirements of section 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The examination and testing of batteries shall be classified as follows:

a. Materials inspection (4.4).

b. In-process inspection (4.5).

c. First article inspection (see 4.6).

d. Quality conformance inspection. (4.7)

1. Inspection of product for delivery (see 4.7.1).

2. Inspection of packaging for delivery (4.7.2)

4.3 Inspection conditions. Except as otherwise specified herein, all examinations and tests shall be performed at a temperature of 80 ± 20°F.

4.4 Materials inspection. Materials inspection shall consist of verifications by certification from the source that the materials used in fabricating the cells and the batteries are in accordance with applicable requirements prior to such fabrication. In the absence of certification from the source, a certificate of analysis or certified inspection data shall be required as proof of conformance to applicable requirements. (see Table I).

TABLE I - MATERIALS INSPECTION

Material	Requirement Paragraph	Applicable Specification
Metals and Finishes Solder (SN 40) Solder Flux (type RMA)	3.4.1	MIL-F-14072 QQ-S-571 MIL-F-14256

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Table II - IN-PROCESS INSPECTION

Inspection	Requirement Paragraph	Test Method Paragraph	Cell	Battery Component	Sampling Plan
Cell closed-circuit voltage	3.6.3	4.8.12	X		100 % of cell lot
Gas leakage	3.8	4.8.13	X		30 cells per cell lot
Low temperature discharge	3.8.1	4.8.13.1	X		4.8.13.I
Coulombic ratio	3.13	4.8.17	X		NA
Cell water content	3.15	4.8.18	X		NA
Insulating Compounds flow and shrinking	3.4.2	4.8.14	X	X	Once per battery lot
Intercell connections	3.5.1	4.8.17		X	4.8.17.1
High temperature switch	3.5.5	4.8.16		X	4.8.16.1

4.5 In-Process Inspection

4.5.2 Cell lot. A cell lot shall be defined as the quantity of cells not exceeding 15,000 cells, manufactured during one month, of any one type and produced at any one place of manufacture on any one contract.

4.5.2.1 Sampling plan. A cell sample lot shall consist of 30 cells per cell lot selected at random.

4.5.3 Battery components. In process inspection on battery components (3.4.2, 3.5.1, and 3.5.5) shall be tested in accordance with Table II.

4.6 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Goverment. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.6.1 Sampling plan. The number of batteries constituting an inspection lot shall be in accordance with Table III.

4.6.2 Inspection routine. First article inspection shall consist of all the examinations and tests in accordance with Table III. One sample battery, untested is to remain at the contractor's plant and is to be

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available as a standard for comparative purposes.

4.6.2.1 First article data. The first article test plan and test report(s) shall be as specified in the contract or purchase order (see 6.2)

4.6.2.2 Failure. If one or more sample batteries fail to meet any of the first article requirements and tests, the contractor shall immediately make the remedial changes. The contractor at no additional cost to the government shall be required to submit additional first article samples for reinspection. A description of the corrective action taken or to be taken shall be included in the first article inspection reports.

TABLE III - FIRST ARTICLE INSPECTION

Group	No. of Batteries	Examination and Test	Requirement Paragraph	Method of Test Paragraph
I	20	Visual & mechanical examination	3.5, 3.13	4.8.1
		Battery voltage	3.6	4.8.2
		Drop test	3.11	4.8.3
		Dimension & weight	3.5	4.8.4
		Mechanical Shock	3.10	4.8.5
		Vibration	3.9	4.8.6
		Altitude	3.12	4.8.7
		Insulation resistance terminals	3.5.3	4.8.8
		Capacity	3.7	4.8.9
IA	10	"I" Test @ 70°F	3.7	4.8.9.1.2
IB	5	"L" Test @-20°F	3.7	4.8.9.1.3
IC	5	"H" Test @ 130°F	3.7	4.8.9.1.4
II	25	Visual-mechanical	3.5, 3.13	4.8.1
		Battery voltage	3.6	4.8.2
		Capacity	3.7	4.8.9
IIA	10	"IT" Test @ 70°F after 4 weeks storage @ 130°F	3.7	4.8.9.1.5
IIB	25 5	"HT" Test @ 130°F after 4 weeks storage @ 130°F	3.7	4.8.9.1.7
IIC	5	"LT"Test @-20°F after 4 weeks storage @ 130°F	3.7	4.8.9.1.6
IID	5	"ILR" Test @ 80°F	3.7	4.8.9.1.10
III	15	Battery Voltage	3.6	4.8.2
IIIA	10	Safety features	3.5.4	4.8.10
IIIB	15 5	Jacket integrity (paperboard)	3.5.6.2.1	4.8.11.2
		Visual-mechanical	3.5, 3.13	4.8.1
IV	1	Untested reference Sample	----	4.6.2

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4.7 Quality conformance inspection.

4.7.1 Inspection of product for delivery. Inspection of product for delivery shall consist of Groups A, B, and C inspection. Test equipment for government verification inspection shall be made available by the contractor, if required.

4.7.1.1 Inspection Battery Lot. A lot shall be defined as the quantity of batteries of any one type produced at any one place of manufacture on any one contract during a one month or less period, submitted at one time to quality conformance inspection.

4.7.1.1.1 Shipment lot. The shipment lot is the quantity of batteries (exclusive of the number of batteries required as samples) of any one type, of any one code, and produced at any one place of manufacture on any one contract on any one contract.

4.7.1.1.2 Contract lot. The contract lot is the total of all batteries (exclusive of the number of batteries required as samples) of any one type, delivered in one or more shipment lots, under the terms of any one contract.

4.7.1.2 Group A. inspection. Each battery on contract or purchase order shall be 100 percent inspected for conformance to the inspections in the order specified in Table IV. Discrete lots shall be formed from batteries that pass this inspection.

TABLE IV GROUP A INSPECTION

Examination and Test	Requirement Paragraph	Method of Test Para.
Visual & mechanical	3.5, 3.13 and table VI	4.8.1
Battery Voltage	3.6	4.8.2

4.7.1.3 Group B inspection. Group B inspection shall consist of the tests specified in Table V in the order shown. Special procedures for small sample inspection of MIL-STD-105 shall apply. Group B inspection shall be performed on sample units form lots which have been subjected to and passed group A inspection.

TABLE V GROUP B INSPECTION

Examination and test	Requirement paragraph	Method of test Paragraph	Inspection AQL Level	
Subgroup I				
Dimensions & Weight	3.5	4.8.1	0.65%	S-1
Insulation resistance	3.5.3	4.8.8	0.65%	S-1
Safety feature	3.5.4	4.8.10	0.65%	S-1

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Table V (cont'd)

Subgroup 2 1/				
Jacket integrity	3.5.6.2.1	4.8.11.2	2.5%	S-1
Visual & mechanical	3.5	4.8.1	2.5%	S-1

1/ Performed only for paperboard jacketed batteries.

TABLE VI - Classification of visual and mechanical examination defects.

Categories	Defects
101	Improper assembly causing parts to be inoperative or unsafe in service.
102	Deformed or damaged parts which are inoperative or malfunction in service.
103	Cell aging requirement not met.
104	Contact surfaces obstructed by insulation material so that electrical use is affected.
105	Torn non-metallic jackets - any tear or rip with dimesion greater than 1/2 inch.
106	Improper jacket closure.
107	Insulating parts or materials missing, damaged, or improperly located so as to affect electrical performance.
108	Location, polarity and marking of terminals not as specified.
109	Labeling and marking wrong, missing or illegible so that utilization is affected.
110	External and internal threads missing, wrong size or so damaged to prevent proper use.
111	Electrolyte leaking caused by missing or defective sealing or closure.
112	Welded or soldered connections improperly made so as to adversely affect battery performance.
113	Improper assembly which could reduce efficiency of operation but not render battery inoperative or unsafe in service.
114	Deformed or damaged parts which do not adversely affect electrical performance.
115	Burrs or imperfections which do not interfere with proper use in operation, assembly or disassembly, or cause unsafe condition in service.
116	Improper marking which doesn't hamper utilization or

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identification of the battery.

4.7.1.4 Group C inspection. Group C inspection shall consist of (a) HT capacity test and (b) LT capacity test. Each capacity test shall be preceded by both the mechanical shock and vibration tests described in paragraphs 4.8.5 and 4.8.6 respectively.

4.7.1.4.1 Sampling plan. Samples shall be selected in accordance with General Inspection Level I from Table I of MIL-STD-105. Fifty percent (50%) shall be used for the HT capacity test and fifty percent (50%) shall be used for the LT capacity test.

4.7.1.4.2 Group C failures. An AQL of 6.5 shall be used for failures (a), (b), and (c). No failures shall be allowed for (d).

a. Insufficient service
b. Excessive initial voltage delay
c. Battery excede dimensions tolerances after discharge.
d. Battery vents, burns or ruptures.

4.7.1.5 Group D inspection. Group D inspection shall be performed at the Government inspection facility on sample batteries in accordance with Table VII. Shipment of the lot represented by the sample batteries shall not be held up pending the results of group D inspection.

TABLE VII - Group D inspection

Storage and Capacity Test	Requirement Paragraph	Method of Test Paragraph
Subgroup I (Adjustment Purposes see 4.7.1.5.3.1)		
Capacity T	3.7	4.8.9.1.7
Capacity LT	3.7	4.8.9.1.5
Capacity ILR	3.7	4.8.9.1.9
Subgroup II (see 4.7.1.5.3.2)		
Capacity D	3.7	4.8.9.1.8

4.7.1.5.1 Sampling plan. Random samples for the Subgroup I capacity tests shall be as specified in Table VIII. Sample size for Subgroup II shall be five (5) batteries per inspection battery lot drawn at random.

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TABLE VIII Sample size and allowable failures for each capacity test under subgroup 1 of group D.

Inspection Battery Lot Size	Sample size for indicated Subgroup I Capacity Test from Inspection Battery Lots			Allowable Failures1/		
	T	LT	ILR	T	LT	ILR
111 to 500 2/	15	15	10	3	3	2
501 to 800	25	25	10	5	5	2
801 to 1300	35	35	10	7	7	2
1301 to 3200	50	50	10	9	9	2
3201 to 8000	75	75	15	13	13	3
8001 to 22,000	110	110	22	18	18	4
22,001 to 110,000	150	150	30	24	24	6
over 110,000	225	225	45	34	34	9

1/ When the number of capacity values falling below the minimum requirements specified (see 3.1) for a given test is equal to or less than the associated allowable failures, the inspection battery lot from which the sample was drawn has met the requirements of that test.

2/ Determination of compliance specified in 4.7.1.5.2 shall not apply to lot sizes of less than 111.

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4.7.1.5.2 Compliance. The entire inspection battery lot shall be considered as complying when the T, LT, and ILR test results show conmpliance as specified in Table VIII.

4.7.1.5.3 Noncompliance.

4.7.1.5.3.1 Subgroup I. If the capacity test results do not show conpliance with the requirements as defined in 4.7.1.5.2, the entire inspection battery lot shall be considered as not complying with requirements of this specification and an adjustment (6.2) can be made if it is determined to be in the Government's best interest by the contracting officer.

4.7.1.5.3.2 Subgroup II. If the number of failures found during capacity test D exceeds twenty (20) percent of the number of samples subjected to this test, the inspection battery lot is considered in noncompliance, and the conditions of paragraph 4.7.1.5.3.1 are applicable.

4.7.2 Quality conformance inspection of packages. Quality conformance inspection shall be in accordance with MIL-P-116. When required (see 6.2) the rough handling requirements shall be in accordance with MIL-P-116.

4.8 Test Methods and examinations.

4.8.1 Visual and mechanical examination. Batteries and cells shall be examined to determine compliance with all applicable requirements and characteristics as specified herein (see 3.5, 3.13, and Table VI.).

4.8.2 Battery Voltage.

4.8.2.1 Open-circuit voltage. A direct current voltmeter of appropriate range and sensitivity shall be used to measure the open-circuit voltage (see 3.6.1). A direct current voltmeter of proper range and sensitivity shall be used to measure the closed circuit voltage utilizing the load specified (see 3.6.2).

4.8.2.2 Closed-circuit voltage. a direct current voltmeter of proper range and sensitivity shall be used to measure the closed circuit voltage utilizing the load specified (see 3.6.2).

4.8.3 Drop test. Each battery shall be dropped once for each temperature, from a height of 30±2 inches onto a hard surface consisting of concrete. The side of the battery perpendicular to and nearest the socket shall be parallel to the concrete surface upon release, and shall make contact with it. The drop test shall be performed at 160°& -20°F. The batteries shall be stabilized a minimum

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of 4 hours at each test temperature and dropped within 10 minutes after removal from the temperature chamber (see 3.11).

4.8.4 Dimensions and weight. Batteries shall be examined by gauging or measuring and by weighing to determine conformance (see 3.5).

4.8.4.1 Dimensions All dimensions shall include any coating which may be used, and shall remain within the specified tolerances throughout the required tests. Both minimum and maximum dimensions shall be determined. When box gauges are used, batteries, loaded with the following weights shall pass freely through the applicable gauge openings:

a. Batteries weighing less than 5 pounds - loading weight of 5 pounds.

b. Batteries weighing 5 pounds or more - loading weight equal to the weight of the battery.

The dimensions of the box guage shall be the specified maximum outside dimensions of the battery.

4.8.5 Mechanical shock. Each battery shall be secured to the testing machine by means of a rigid mount which will support all mounting surfaces of the battery. Each battery shall be subjected to a total of three shocks of equal magnitude. The shocks shall be applied in each of three mutually perpendicular directions. Each shock shall be applied in a direction normal to a face of the battery. The faces of the battery are identified by their position in relation to the front face (the face which bears the label or marking). For each shock, the battery shall be accelerated in such a manner that during the first 3 milliseconds the minimum average acceleration is 75 gravity units (G). The peak acceleration shall be between 125 to 175G, exact value shall be recorded (see 3.10).

4.8.6 Vibration. Each battery shall be rigidly clamped to the platform of a vibration machine in a manner approximating as closely as practicable the manner in which the batteries are clamped when in use (see 3.1). A simple harmonic motion shall be applied having an amplitude of 0.03 inch (0.06 - inch total maximum excursion). The frequency shall be varied at the rate of 1 hertz per minute between the limits of 10 and 55 cycles per second. The entire range of frequencies and return shall be traversed in 95±5 minutes for each mounting position, direction of vibration) of the battery. The batteries shall be vibrated in three equal periods in mutually perpendicular directions, one of which shall be perpendicular to the terminal face of the battery. Open-circuit voltage shall be observed for 30 seconds during the last quarter of each of the three vibration periods (see 3.9).

4.8.7 Altitude Batteries shall be placed in an altitude chamber, in which the pressure is maintained at a value corresponding to an altitude of 50,000 feet and the temperature is kept at 75±5° for a period of six hours (see 3.12).

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4.8.8 Insulation resistance (terminals). Insulation-resistance test shall be performed, except as otherwise specified (see 3.1). Batteries shall be stored for a period of 48 hours at +70±5°F for a relative humidity of 50±15 percent. After storage and while at these conditions, the insulation resistance shall be measured by applying a direct-current potential of 500±20 volts between any two terminals not electrically connected and between all ungrounded terminals and the container of the battery. The insulation resistance of batteries having a non-metallic container shall be measuired by the use of a 1 inch-square copper plate making physical contact with the container . The plate shall be placed with the broad surface against any area of any surface of the jacket other than that on which the battery terminals are located (see 3.5.3).

4.8.9 Capacity(see 3.7).

4.8.9.1 Capacity Tests. Sample batteries selected for capacity tests specified in the individual specification sheet (3.1) shall be stored and discharged in air as applicable in accordance with 4.8.9.1.1 through 4.8.9.2.1. All batteries shall be discharged to zero volts, except those subjected to the ILR test. The time required to reach the specified minimum voltage shall be used to determine the battery capacity.

4.8.9.1.1 Initial voltage delay. At the start of the capacity discharge test, batteries shall be monitored with an oscillographic device or equivalent to determine the time in milliseconds required for battery units to reach minimum voltage after the specified loads are applied as specified in the individual specifications sheets. No loads shall be applied at any time during storage or prior to the discharge test following such storage (see 3.7.1).

4.8.9.1.2 Capacity test I. Discharge at +70±5°F without previous storage. After the battery reaches zero volts, force discharge the battery into voltage reversed for the specified time.

4.8.9.1.3 Capacity test L. Discharge at -20±3°F after storage at -20±3°F for a minimum of sixteen (16) hours.

4.8.9.1.4 Capacity test H. Discharge at +130±3°F after storage at +130 F±3°F for a minimum of sixteen (16) hours.

4.8.9.1.5 Capacity Test IT. Discharge at +70° ± 5°F after four (4) weeks storage at 130°F. After the battery reaches zero volts, force discharge the battery into voltage reversal for the specified time.

4.8.9.1.6 Capacity test LT. Discharge at -20±3°F after four (4) weeks storage at 130°F followed by a minimum of sixteen (16) hours at -20±3°F. No load shall be applied during the storage at 130°F and -20°F or prior to discharge test.

4.8.9.1.7 Capacity test HT. Discharge at ±130±3°F after four (4) weeks storage at 130°F and a minimum of sixteen (16) hours at +130±3°F. No load shall be applied during the storage at 130°F and 130°F or prior to discharge test.

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4.8.9.1.8 Capacity test T . Discharge at +70±5°F after thirteen (13) weeks storage at +113°F.

4.8.9.1.9 Capacity test D . Discharge at +70±°F after fifty-two (52) weeks storage at +70°F.

4.8.9.1.10Capacity test ILR. Discharge at +80±20°F to the specified minimum voltage without previous storage.

4.8.9.2 Storage conditions The storage conditions specified in Table IX shall prevail during storage periods specified. Normal conditions shall be maintained. No loads shall be applied at any time during storage or prior to the discharge test following storage. Deviations from normal conditions are permitted provided that (1) the extreme conditions specified in Table IX do not exist for more than ten percent (cumulative of the specified storage periods) and (2) that at no time are the extreme conditions exceeded. Batteries must be oriented in storage to have at least 50 percent of the cell seals at the bottom of the batteries (see 3.1).

Table IX Storage test conditions

Kinds of storage	Normal conditions temperature	Extreme Conditions temperature
HT and LT Storage	+130° + 3° F - 4° F	108°F thru 126°F and 133°F thru 135°F
D Storage	+70° ± 5°F	60°F thru 65°F and 75°F thru 80°F
T Storage	+113°+ 3°F - 7°F	

4.8.9.2.1 Discharge. Following stabilization the batteries shall be discharged at the ambient discharge conditions as specified. There shall be at least a two (2) inch separation between all batteries that are being discharged. All batteries subjected to capacity testings shall be discharged as specified in individual specification sheets. A battery shall be considered to have failed a discharge test when any one of the following conditions occurs:

a. The battery voltage or the voltage of any one unit falls below the specified minimum voltage prior to exceeding the capacity or fails to reach minimum voltage in the required time.

b. The battery dimensions exceed the maximum specified (see 3.1).

c. Any evidence of cell venting, rupture or fire is observed.

d. Insufficient service.

4.8.10 Safety feature Each sample battery shall be subjected to a

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batteries shall be placed at +195°F for two hours. At the end of the two hours the battery must be checked to see that no cell has vented. If any cell has vented, the battery has failed this test. If no cell has vented, the ambient temperature shall be raised to a maximum of 320°F. Batteries shall be maintained at an ambient of 320° F for two hours. Any cell that vented at or below 320°F must be checked to insure that the venting occurred only through the designed vent. Venting through any portion of the cell other than the designed vent is a failure. Batteries containing cells which exploded, burned or did not vent are failures. (see 3.5.4) Distortion of labeling or dimensions is permitted.

4.8.11 Jackets.

4.8.11.1 Metallic jackets. Metallic jacketed batteries weighing five pounds or more shall be loaded by applying weights totaling 100 pounds evenly distributed over the top of the battery and shall remain so loaded at least one minute. The condition of the jacket shall be observed (see 3.5.6).

4.8.11.2 Jacket integrity (paperboard). Batteries shall be immersed to within 1/4 inch of the top of the jacket in water maintained at a temperatue of +113±5°F for a period of 48 hours. The condition of the jacket and its seams shall be observed (see 3.5.6.2.1). This test is only to be conducted on paperboard jackets.

4.8.12 Cell lot test. Closed-circuit voltage test. All the cells in a cell lot shall be pulse tested for five (5) seconds at 0.250 amperes. Cell test capacity, C, shall be defined as the capacity in ampere hours to 2.5 at the 24 hour rate. Any cell whose voltage falls below 2.5 volts during the 5 second pulse period shall be rejected for further cell testing (see 3.6.3).

4.8.13 Gas leakage test. At no time shall there be potting substances or cell jackets applied to the cells designed for this test. Each of the thirty (30) cells shall be weighed prior to and after filling with the electrolyte thionyl chloride mixture. Each cell shall be weighed to the nearest tenth of a milligram and the weight recorded. The amount of thionyl chloride shall be determined for each cell based on the percentage used. The cells shall then be stored for one week (7 days) at 130°F. At the end of one week, the cells shall be removed from the temperature cabinet, placed in a desiccator, and cooled at room temperature for at least two hours. Each cell shall be weighed to the nearest tenth of a milligram. All cells shall be placed in the temperature cabinet and stored for three weeks (21 days) at 130°F. At the completion of this three week storage period, the cells shall be removed, placed in a desiccator and cooled for at least two hours at room temperature. Each cell shall be weighed to the nearest tenth of a milligram. The weight loss between day 7 and day 28 shall be recorded (see 3.8). The weight loss shall not exceed 0.005 percent of the total thionyl chloride weight, nor shall there be any corrosion evident. The solvent and the electrolyte salt weights shall not be used in this computation. If there are one or more failures the cell lot is rejected.

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4.8.13.1 Low temperature discharge test. Following the gas leakage test, the same thirty (30) cells used in the gas leakage test shall be stored at -20° ± 3°F for a minimum of six (6) hours. Each cell shall be discharged at -20° ± 3°F through the load and to end voltage in accordance with the individual specification sheet. There shall be at least a two (2) inch separation between all cells that are being discharged. Following capacity determination, the discharge shall continue to zero volts. Capacity shall be no less than 75% of the battery capacity listed for the low temperature (LT) test on the applicable battery specification sheet.

a. If all thirty (30) cell samples meet the requirements of 3.8.1. the cell lot may be used for fabrication of batteries.

b. If there are two (2) or more failures during the capacity testing of the thirty (30) cell sample, the cell lot shall be rejected. If only one (1) failure occurs, another twenty (20) cell sample shall be selected from the same lot and this same test shall be repeated. If no failures occur during retesting, the cell lot shall be considered acceptable. If one (1) or more cells fail during retesting, the cell lot shall be rejected and a new lot of cells shall be submitted for cell lot inspection.

4.8.14 Flow or shrinking (insulating compounds) (see 3.4.2). Compounds shall be placed in a container, approximately 3 inches wide by 6 inches long by 3/4 inch high, to within 1/4 inch of the top. The temperature of the compound within the container shall be raised to +200±5°F and the container shall be held in an inverted position for 24 hours. Then the temperature of the compound shall be lowered to -40±5°F.

4.8.15 Connector location. Connector location shall be verified by use of a mating connector.

4.8.16 High temperature switch test. This test shall be performed on switches or thermal fuses prior to installation in batteries. Test sample shall be placed in a temperature chamber at +175±5°F for a minimum of two hours. Each sample shall then be checked to verify that the switch or thermal fuse is closed. The temperature shall be raised to +190±5°F. After 45 minutes, each sample shall be checked to verify that th e switch or thermal fuses is open. (see 3.5.5)

4.8.16.1 Sample size. The number of high temperature switches or thermal fuses tested per inspection battery lot shall equal the same number of batteries subjected to the "T" capacity test per inspection battery lot.

4.8.16.2 Failure rate. The failure rate shall be the same as for the "T" capacity test in Table VIII.

4.8.17 Coulombic ratio. Coulombic ration for lithium to thionyl chloride shall be certified for first article imspection in the first article test report and for each inspection battery lot in the quality conformance test report (see 3.13 and 6.2).

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4.8.18 Intercell connections. Intercell connections shall be visually examined for conformance to 3.5.1.

4.8.18.1 Sample size. The number of batteries examined per inspeciton battery lot shall equal the same number of batteries subjected to the "T" capacity test per inspection battery lot.

4.8.18.2 Failure rate. The failure rate shall be the same as for the "T" capacity test in Table III.

4.8.19 Cell water content Cell water content shall be certified for first article inspection in the first article test report and for each inspecion battery lot in the quality conformance test report. (see 3.14 and 6.2).

5 PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI 1G0066.

5.2 Shipments. Department of Transportation DOT-E-7052 or applicable CFR 49 in effect at the time of shipments shall apply. Batteries shall not require metal drums as the outside shipping container for air shipment.

6 NOTES

6.1 Intended use. The primary batteries included are of the non-reserve type composed of electrochemical cells utilizing a lithium thionyl chloride system. The lithium cell batteries are capable of use under wide temperature ranges. Storage above 130°F should be limited.

6.1.1 Indirect shipments. The packaging, packing and marking specified in Section 5 apply only to direct purchases by or direct shipment to the Government and are not intended to apply to contracts or orders between the supplier and prime contractor.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Applicable specification sheet (see Supplement 1).

c. Complete type designation (see 1.2.1).

d. Requirement for first article test plan and test report (see 4.6.2.1)

e. Requirement for maintaining manufacturing processes (see 3.5.8)

f. Requirement for certification of materials and processes (see 3.4).

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g. Requirement for certification of coulombic ratio (see 3.13).

h. Requirement for certification of age of cells (see 3.5.2).

i. Requirement for contractor to state color of jackets (see 3.5.6.3).

j. Adjustment procedures for Group D inspection for Capacity (see 4.7.1.5.3).

k. If rough handling test is required (see 4.7.2).

l. Requirement for quality conformance test report (see 3.13 and 4.8.17).

m. Requirement for certification of cell water content (see 3.15).

6.3 Definitions

6.3.1 Non-flammable and non-toxic materials. Non-flammable and non-toxic materials are those materials which will not support combustion, produce smoke, or be capable of emitting toxic fumes when subjected to the environmental conditions specified for the battery.

6.4 International standardization agreement. Provision of this specification are the subject of international standardization agreements. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standarization Offices so that appropriate action may be taken respecting the international agreement concerned.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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